ChronoSafe-Drive

Self-Diagnosis, Risk Calibration, and Minimal-Risk Maneuvers under Multi-Sensor Perception Degradation

AMD Hackathon 2026 - Track 3: Physical AI

Competition Technical Report

Closed-loop MRM	Timestamp robustness	AMD ROCm inference
20/20 success 0% premature	**95.08%-99.34% stress accuracy**	**0.559 ms p50 1.37M seq/s**

Platform: AMD Radeon gfx1100 | ROCm/HIP 7.2 | PyTorch 2.9.1 ROCm

Evaluation: nuScenes v1.0-trainval + 20 unseen MetaDrive closed-loop seeds

Wang Zhenyu | Wang Guanyu | Han Liuxuan

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Abstract

ChronoSafe-Drive is an AMD Radeon and ROCm Physical AI system for autonomous driving under persistent multimodal perception degradation. The system combines physics-based MetaDrive simulation with public nuScenes camera/LiDAR streams, extracts 15 causal health and timing features, and applies an eight-step factorized GRU to diagnose camera, LiDAR, and temporal faults while predicting continuous driving risk. Timestamp-authenticity augmentation exposes the model to missing, drifting, jittered, swapped, and spoofed metadata. A stress-aware split-conformal procedure converts risk predictions into conservative upper bounds, which drive a frozen three-state minimal-risk controller. On 20 unseen MetaDrive seeds, the system reaches a minimal-risk condition in every run with zero premature triggers, 97.0714% mean scenario accuracy, and 99.8333% mean upper-risk coverage. On held-out nuScenes scenes, timestamp-robust training improves severe corruption accuracy by 11.38 to 27.32 percentage points while retaining 99.3446% intact accuracy. AMD ROCm inference is sub-millisecond and reaches 1.37 million temporal sequences per second at batch 1024. The complete pipeline demonstrates multimodal perception, robustness, calibrated decision making, GPU acceleration, and closed-loop physical control.

1. Problem statement

Autonomous-driving stacks assume that camera images, LiDAR measurements, and their timestamps describe the same current world state. This assumption can fail gradually or persistently. Images may blur or freeze; LiDAR returns may drop out; delivery delay may grow; and timing metadata may be missing or incorrect. A detector trained only on isolated faults may also fail when two modalities degrade together.

ChronoSafe-Drive studies the following question:

> *Can a causal, multimodal temporal model diagnose persistent and compound perception degradation, estimate a statistically calibrated risk upper bound, and trigger a minimal-risk maneuver without premature stopping?*

The project is designed around the complete Physical AI loop rather than an offline classifier alone: perception evidence changes diagnosis, diagnosis changes calibrated risk, risk changes the driving mode, and the resulting control action changes the simulated vehicle state.

1.1 Target application and practical value

The target application is a safety supervisor for multimodal autonomous-driving systems operating in complex environments. It is not intended to replace the primary perception or planning stack. Instead, it monitors whether camera, LiDAR, and packet-timing evidence remain trustworthy and requests reduced-speed or minimal-risk behavior before persistent degradation can propagate into unsafe control.

The same architecture is applicable to autonomous shuttles, delivery vehicles, industrial mobile robots, and other Physical AI systems that must remain safe during partial sensor, synchronization, or communication failure. Its practical value is the integration of diagnosis, calibrated uncertainty, and physical control rather than a fault classifier that stops at an offline metric.

2. Contributions

1. **Factorized temporal diagnosis.** A shared GRU models temporal health while independent camera, LiDAR, and timing heads preserve the compositional structure of multi-sensor degradation.
2. **Timestamp-authenticity augmentation.** The training protocol corrupts the metadata channel itself, including the difficult case where stale packets are marked as current, and calibrates against the worst timestamp-stress domain.
3. **Calibrated minimal-risk control.** A split-conformal upper risk bound feeds a hysteretic state machine with a calibration-only threshold and a latched stop state.
4. **Leakage-resistant real-data protocol.** All derived tracks from one nuScenes scene remain together, and adjacent scenes are grouped by complete `log_token` before train/calibration/test assignment.
5. **Reproducible AMD ROCm evidence.** Training, evaluation, AMP verification, and deployment-style inference are executed on an AMD Radeon `gfx1100` GPU using PyTorch ROCm/HIP 7.2.

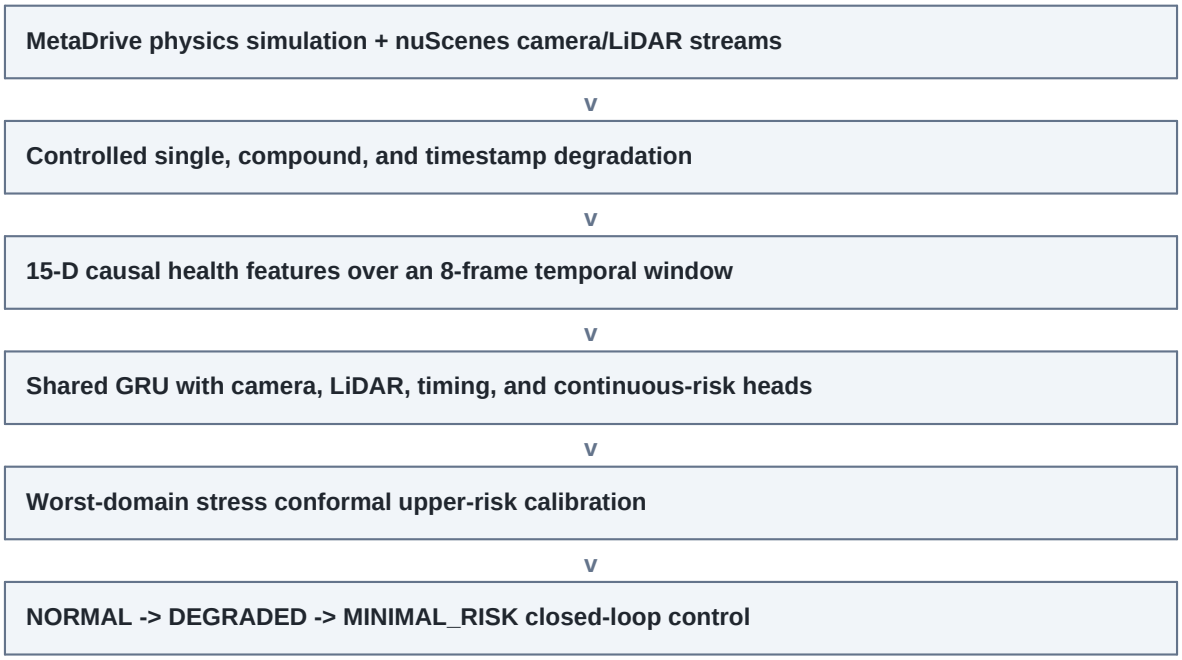
2.1 Team structure and individual contribution

ChronoSafe-Drive is a three-member team submission with the following declared contributions:

- **Wang Zhenyu — Team Lead and Primary Developer:** problem definition, system architecture, degradation design, dataset construction, model and controller implementation, AMD ROCm training and benchmarking, calibration protocol, experiments, statistical evaluation, visualization, documentation integration, and technical submission delivery.
- **Wang Guanyu — Demo Video Producer:** demo planning, screen recording, editing, English subtitles, and final video production.
- **Han Liuxuan — Research Materials Coordinator:** collection, organization, and verification of competition, technical, and reference materials used to support the project and submission.

Except for the demo-video production and research-material coordination roles listed above, the remaining project work was completed by Wang Zhenyu. Third-party open-source frameworks and datasets are identified in the repository documentation and remain subject to their own licenses.

3. System overview



4. Multimodal degradation model

4.1 Controlled perception conditions

Seven scenario labels are used consistently across the simulator and real sensor streams:

1. clean;
2. progressive camera blur;
3. camera freeze;
4. LiDAR dropout;
5. dynamic cross-modal delay;
6. camera freeze plus LiDAR dropout; and
7. camera blur plus dynamic delay.

The continuous target risk is a clipped weighted sum of observable controlled severity:

```
r = clip(0.25 * blur + 0.30 * freeze + 0.35 * dropout + 0.20 * delay, 0, 1)
```

Here, blur, dropout, and delay are normalized to their configured maximum severity. This target defines the controlled experimental risk and is not presented as a certified real-world collision probability.

4.2 Timestamp stress conditions

Training applies timestamp augmentation to 60% of sampled sequences. A selected sequence receives one of the following conditions:

- a 30% missing-timestamp burst;
- three-frame progressive camera clock drift;
- one-frame random timestamp jitter;
- swapped camera and LiDAR timestamp channels; or
- timestamps spoofed to appear clean.

Stress calibration and final testing use stronger missing, drift, and jitter levels. Corruption is applied only to causal timestamp features; ground-truth labels and risk targets remain unchanged.

5. Causal feature representation

No clean reference stream is available to the deployed model. At each step, the extractor produces:

- camera mean, standard deviation, log sharpness, and temporal change;
- LiDAR mean, standard deviation, minimum range, zero ratio, and temporal change;
- normalized ego speed;
- normalized camera packet age and LiDAR packet age;
- cross-modal packet-age skew;
- cross-modal temporal-change disagreement; and
- the temporal change in LiDAR zero ratio.

The resulting feature vector has dimension 15. Eight consecutive vectors form one causal model input with shape `[8, 15]`.

6. Factorized temporal model

For an input window `x[t-7:t]`, the shared GRU produces a final hidden state:

```
h_t = LayerNorm(GRU(x[t-7:t]))
```

Four prediction heads operate on `h_t`:

- camera state: normal, blur, or freeze;
- LiDAR state: normal or dropout;
- timing state: synchronized or delayed; and
- continuous risk in `[0, 1]`.

The three factor predictions are composed deterministically into the seven evaluation scenarios. The final robust model contains 56,968 parameters, uses hidden size 128, and is trained with factor cross-entropy and risk regression. The direct seven-class scenario-loss weight is fixed to zero in the selected robust variant to retain factor compositionality.

Training mixes 60,000 MetaDrive sequences with 60,000 nuScenes sequences per run, uses batch size 512 for 100 epochs, and repeats the frozen protocol with random seeds `2026`, `2027`, and `2028`.

7. Stress-aware conformal risk calibration

Let `r_i` be the controlled target risk and `r_hat_i` the predicted risk for a calibration sample. The nonconformity score is:

```
e_i = abs(r_i - r_hat_i)
```

For target miscoverage `alpha = 0.1`, a finite-sample conformal quantile is computed independently for each held-out calibration domain and timestamp stress condition. The deployed radius is the maximum selected radius:

```
q_stress = max_domain,condition Quantile_0.9(e)
upper_risk = min(1, r_hat + q_stress)
```

This deliberately conservative selection protects against the strongest calibration-time timestamp condition. Test domains never participate in radius or threshold selection.

8. Minimal-risk maneuver controller

The control policy exposes three interpretable modes:

- `NORMAL`: nominal steering and speed control;
- `DEGRADED`: reduced target speed and conservative commands; and
- `MINIMAL_RISK`: latched braking until the vehicle reaches a near-stop.

An eight-step warm-up prevents decisions before a full temporal window exists. Hysteresis prevents rapid mode switching. The minimal-risk threshold is selected from only MetaDrive calibration seeds `113` and `127`; candidate thresholds are evaluated against success, premature triggering, and delay. Threshold `0.62` achieves 100% calibration success, 0% premature triggers, and the lowest valid mean trigger delay, and is frozen before external testing.

9. Data and leakage prevention

9.1 nuScenes trainval

The public nuScenes v1.0-trainval conversion contains:

- 850 scenes;
- 34,149 aligned `CAM_FRONT` + `LIDAR_TOP` pairs;
- 239,043 degradation samples;
- 5,950 temporal tracks; and
- 15 features per step.

The split contains 592 training, 128 calibration, and 130 test scenes. The unit of grouping is the complete nuScenes `log_token`, which prevents adjacent driving snippets from the same collection log from appearing in different partitions. All augmentations from one scene remain in the same partition.

9.2 MetaDrive

MetaDrive provides physical vehicle dynamics, camera rendering, LiDAR returns, and closed-loop actuation. Training, calibration, internal test, and external closed-loop seeds are disjoint. Twenty external seeds are frozen in the protocol and reported without outcome-based selection.

9.3 Renderer integrity

MetaDrive 0.4.3 can fail in a display-less environment during SimplePBR setup. Bypassing the camera effects allows execution but changes the camera features by more than 20 standard deviations and invalidates the closed-loop diagnosis result. The official evaluation therefore uses `xvfb-run`, disables the global render pipeline consistently, and leaves the camera-effect code unmodified. A three-seed probe confirms exact 15-feature equality with the original training data (`mean_abs_diff = 0`, `shift_sd = 0`). All raw-headless bypass results are excluded from claims.

10. Experimental results

10.1 Timestamp robustness

Each condition contains 21,973 temporal sequences from the same leakage-controlled 95-scene nuScenes test subset. Both model variants are summarized over three training seeds; the robust column reports mean +/- standard deviation across those seeds.

Condition	Standard v3	Robust v3 mean +/- SD	Gain
Intact	99.8271%	99.3446% +/- 0.1093%	-0.4825 pp
Missing 50%	84.7828%	97.7715% +/- 0.0730%	+12.9887 pp
Jitter, two frames	85.3047%	96.6868% +/- 0.4945%	+11.3821 pp
Timestamp channels swapped	72.0202%	99.3446% +/- 0.1093%	+27.3244 pp
Metadata spoofed clean	67.9637%	95.0818% +/- 0.0682%	+27.1181 pp

The result supports the central claim: corrupting timestamp metadata during training substantially improves robustness to persistent metadata failure while reducing intact accuracy by less than half a percentage point.

10.2 Fair nuScenes comparison

This comparison uses 95 scenes grouped into 8 independent `log_token` units; every log containing a nuScenes-mini scene is excluded. Confidence intervals use 10,000 paired log-cluster bootstrap resamples.

Metric	Locked v2	Robust v3	Relative or absolute improvement
Scenario accuracy	80.5762%	99.3446%	+18.7684 pp
Camera accuracy	94.9438%	99.7816%	+4.8378 pp
LiDAR accuracy	85.7097%	99.9985%	+14.2888 pp
Risk MAE	0.060401	0.012068	-80.02%
Conformal radius	0.251078	0.058434	-76.73%

The standard v3 model reaches higher intact accuracy and lower risk MAE than the robust v3 model. The final selection is therefore justified by the frozen robustness objective and safety coverage, not by claiming dominance on every clean-input metric.

10.3 External closed-loop evaluation

Metric over 20 unseen seeds	Mean +/- SD or rate
MRM success rate	100%
MRM reached rate	100%
Premature MRM rate	0%
Scenario accuracy	97.0714% +/- 2.8541%
Risk MAE	0.020637 +/- 0.003302
Upper-risk coverage	99.8333% +/- 0.2796%
Trigger delay	14.05 +/- 2.11 steps
Braking time	8.55 +/- 3.15 steps
Final speed	0.013786 +/- 0.002101 km/h

The controller spends 53.2619% of steps in `NORMAL`, 24.8571% in `DEGRADED`, and 21.8810% in `MINIMAL_RISK`, demonstrating a complete staged transition rather than an immediate always-stop policy.

11. AMD Radeon and ROCm implementation

11.1 Platform

- AMD Radeon Graphics, `gfx1100`, 48 compute units;
- 47.984 GiB VRAM;
- HIP `7.2.53211`;
- PyTorch `2.9.1+rocm7.2.1`; and
- Linux 6.8, Python 3.12.

Training and inference both require `torch.cuda.is_available()` and a non-null `torch.version.hip`; the scripts fail instead of silently falling back to CPU.

11.2 Real-sequence inference benchmark

The benchmark uses 30,142 frozen test sequences across 130 nuScenes scenes. After 50 warm-up iterations, each precision/batch configuration is timed for 300 synchronized ROCm GPU-event iterations.

Precision	Batch	p50	p95	p99	Throughput	Peak allocated
FP32	1	0.558821 ms	0.954580 ms	1.019299 ms	1,632.89 seq/s	77.312 MiB
FP32	32	0.571206 ms	0.974368 ms	1.056337 ms	51,199.80 seq/s	77.794 MiB
FP32	256	0.636986 ms	1.010295 ms	1.078844 ms	367,947.68 seq/s	82.494 MiB
FP32	1024	0.714952 ms	1.003659 ms	1.064934 ms	1,371,235.75 seq/s	98.604 MiB
AMP FP16	1024	0.694567 ms	1.086294 ms	1.201812 ms	1,364,850.70 seq/s	88.600 MiB

Batching increases FP32 throughput by approximately 840x while p50 latency increases by only 1.28x. At batch 1024, AMP improves p50 by 1.029x and saves approximately 10 MiB of peak allocated memory. Small-batch AMP is slower because the compact recurrent model is launch-bound; this negative result is preserved in the machine-readable benchmark rather than omitted.

AMP numerical verification uses 8,192 real test sequences. All three diagnostic heads achieve 100% class agreement with FP32. Risk prediction MAE is `0.00008262`; p99 error is `0.00033322`; maximum error is `0.00064528`.

12. Reproducibility

The repository preserves:

- frozen seed and split manifests;
- deterministic training options;
- SHA-256 hashes of benchmarked data, split, and checkpoint;
- JSON/CSV per-run and aggregate metrics;
- PNG figures with the underlying machine-readable tables;
- calibration-only MRM threshold selection;
- a dependency-free verifier that checks public claims against frozen JSON; and
- explicit hardware, HIP, PyTorch, and timing metadata.

Exact commands are in the root `README.md`. Raw media and generated datasets are excluded; generation instructions, small checkpoints, and the split manifest remain at their documented reviewer paths.

13. Final deliverables and output formats

The competition submission is organized as a dedicated public source repository and contains the following deliverables:

Deliverable	Format	Purpose
Project source code	Python package and executable scripts	Data construction, training, evaluation, benchmarking, and closed-loop control
Technical report	Markdown	Target application, architecture, data, AMD use, innovations, results, and limitations
Reproducibility guide	Markdown plus pinned dependency file	Environment setup and step-by-step result reproduction
Frozen protocol	JSON and Markdown	Seeds, splits, calibration policy, and statistical rules
Model artifacts	PyTorch checkpoints with SHA-256 records	Reproduce the selected model without retraining
Quantitative evidence	JSON, CSV, Markdown, and PNG	Machine-readable metrics and reviewer-facing figures
Closed-loop demonstration	Public English video no longer than 3 minutes	Show commands or interface actions, execution, mode transitions, and results
Dataset documentation	Markdown and generation scripts	Provenance, licensing constraints, split policy, and regeneration

The submission excludes raw nuScenes media and other large licensed data. These are obtained from their official source and converted with the included scripts. Reviewer-facing commands use the persistent AMD cloud workspace path and fail explicitly when the required ROCm/HIP device is absent.

14. Limitations and responsible use

1. The evaluated faults are controlled corruptions, not naturally annotated hardware failures.
2. nuScenes supports offline sensor evaluation; physical control is evaluated in MetaDrive rather than on a road vehicle.
3. The 15-dimensional representation prioritizes explainability and efficient temporal health monitoring over end-to-end visual semantics.

4 Upper-risk coverage is empirical under the frozen domains and does not imply universal safety certification.

5 The controller is a research prototype and must not be deployed as the sole safety mechanism in a real vehicle.

All users must comply with nuScenes licensing terms and applicable privacy, legal, and ethical requirements. The repository does not redistribute raw nuScenes images or point clouds.

15. Conclusion

ChronoSafe-Drive demonstrates that temporal sensor-health diagnosis becomes substantially more robust when timestamp metadata is treated as a fallible input rather than trusted ground truth. Combining factorized diagnosis, stress-aware risk calibration, and a frozen minimal-risk controller produces a transparent closed loop that succeeds on every unseen MetaDrive test seed. The system is trained and evaluated on AMD Radeon and ROCm, with reproducible sub-millisecond inference, numerical AMP verification, and complete machine-readable evidence.

AMD ROCm benchmark figure

